

AI-Agent Resume Screening

Transforming recruitment efficiency with intelligent automation



The Hiring Challenge

Modern HR teams face unprecedented volume and complexity in resume screening that traditional processes cannot efficiently handle.

Volume Overload

1,000+ resumes per role create overwhelming workloads for recruitment teams

Inconsistent Standards

Manual review leads to subjective evaluation and lack of standardized criteria

Missed Opportunities

Qualified candidates overlooked due to time constraints and evaluation fatigue

Delayed Communication

Slow response times frustrate candidates and damage employer brand

Introducing the Agentic AI Solution

An automated, intelligent hiring workflow that combines n8n automation with OpenAI GPT models to revolutionize resume screening.

01

Workflow Automation

n8n orchestrates end-to-end screening process

02

AI Intelligence Layer

OpenAI GPT analyzes and evaluates candidates

03

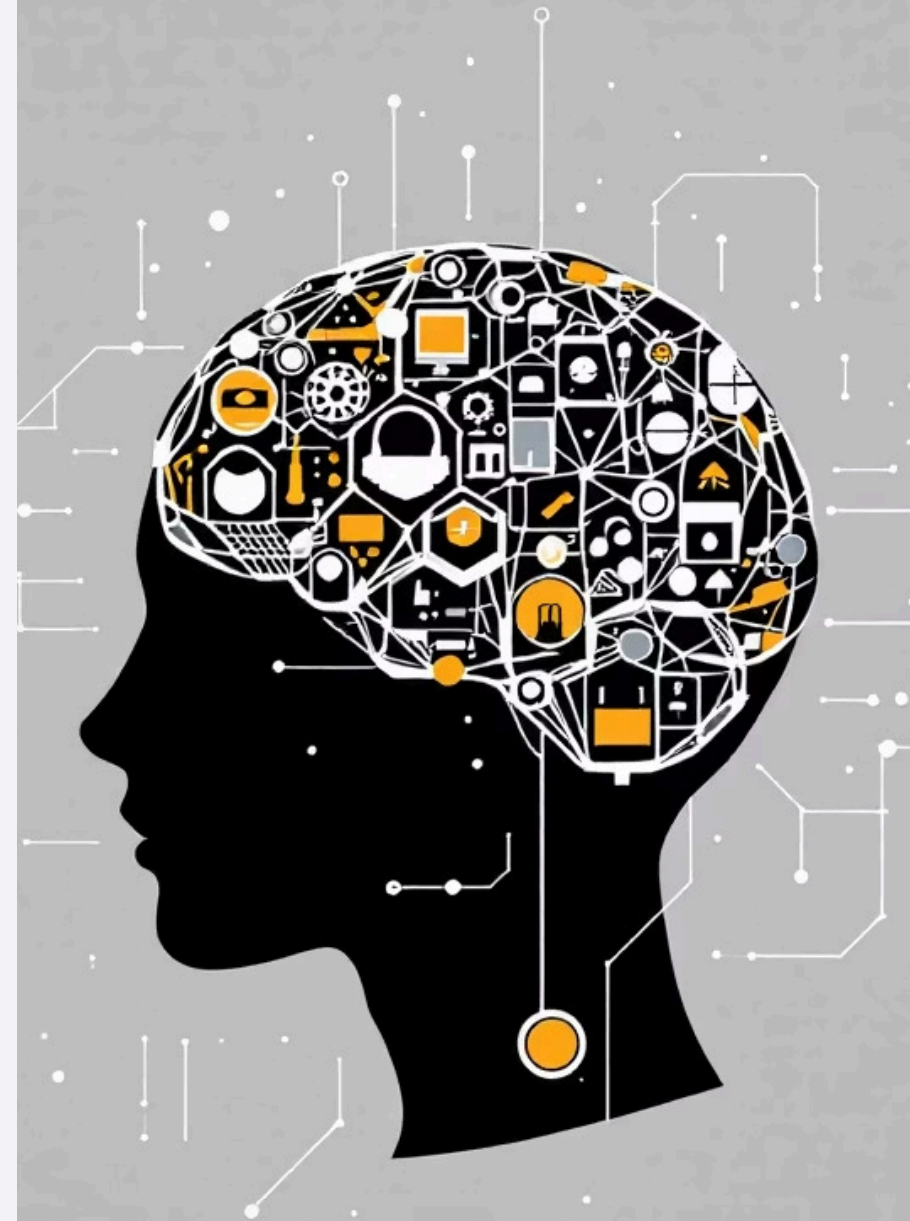
Structured Output

JSON parsing ensures clean, actionable data

04

Decision Logic

Automated shortlist and rejection workflows



AI Decision Framework

How AI Evaluates Candidates

The system leverages advanced artificial intelligence to extract, analyze, and score candidate qualifications against job requirements.

Decision Threshold

Match score $\geq 70 \rightarrow$ **Shortlisted**

Match score $< 70 \rightarrow$ **Rejected**

Fair, consistent, and bias-minimized screening ensures quality while maintaining equity.

Structured Data Extraction

Parses resume content into standardized fields

Experience Assessment

Calculates years and relevance of background

Skill Match Analysis

Identifies and weights relevant competencies

Job Description Alignment

Measures fit against role requirements

Complete Workflow Process



Submission

Candidate uploads resume through form



Extraction

System extracts and normalizes text



Evaluation

AI analyzes and scores match



Storage

Database logs all data and decisions



Communication

Automated email notifications sent

Workflow in Detail

1

Resume Submission

Candidates upload through integrated form or trigger mechanism

2

Text Extraction & Normalization

Content cleaned and structured for consistent AI processing

3

AI Processing (OpenAI GPT)

Model extracts details, identifies skills, calculates matchScore, generates recommendations

4

JSON Parsing

Converts raw AI output into structured fields: name, email, phone, experience, skills, score, recommendation

5

Decision Logic

Automated condition applies passing score threshold of 70 for shortlist or rejection

6

Data Storage & Audit

Complete traceability with timestamps and logged decisions for transparency

Operational Benefits



Time Savings

70–85% reduction in manual screening time per candidate



Bulk Processing

Handles 1,000+ resumes efficiently in single batch operations



Fully Automated

Eliminates manual decision-making for initial screening stages



Standardized Framework

Consistent evaluation criteria applied uniformly across all candidates



Workload Reduction

Free up recruiters for strategic activities and candidate engagement



Productivity Boost

Doubles recruiter effectiveness through intelligent automation

Talent & HR Impact



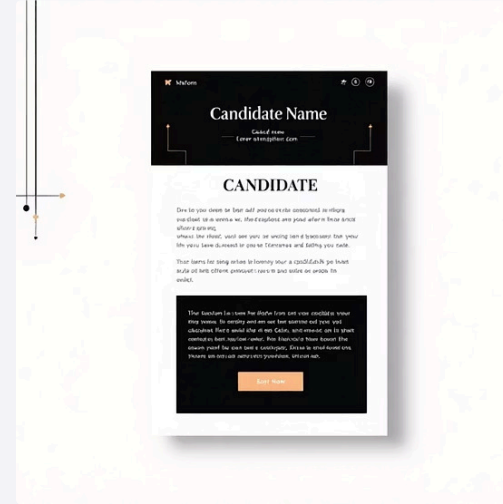
Faster Hiring Cycles

3–5x acceleration from application to shortlist enables competitive talent acquisition



Better Quality Shortlists

Data-driven scoring identifies strongest candidates while reducing subjective bias



Improved Candidate Experience

Faster response times and professional communication enhance employer brand perception



Structured Audit Trail

Complete traceability supports compliance, reporting, and continuous process improvement

Business Impact Summary

80%

Time Saved

Reduction in HR
operational effort per
hiring cycle

60%

Cost Reduction

Lower cost per hire
through automation
efficiency

3X

Faster Hiring

Accelerated
shortlisting and
interview scheduling

Unli...

Scalability

No limit on volume
without additional
headcount

Strategic Advantages

- **Scalable Hiring Model**

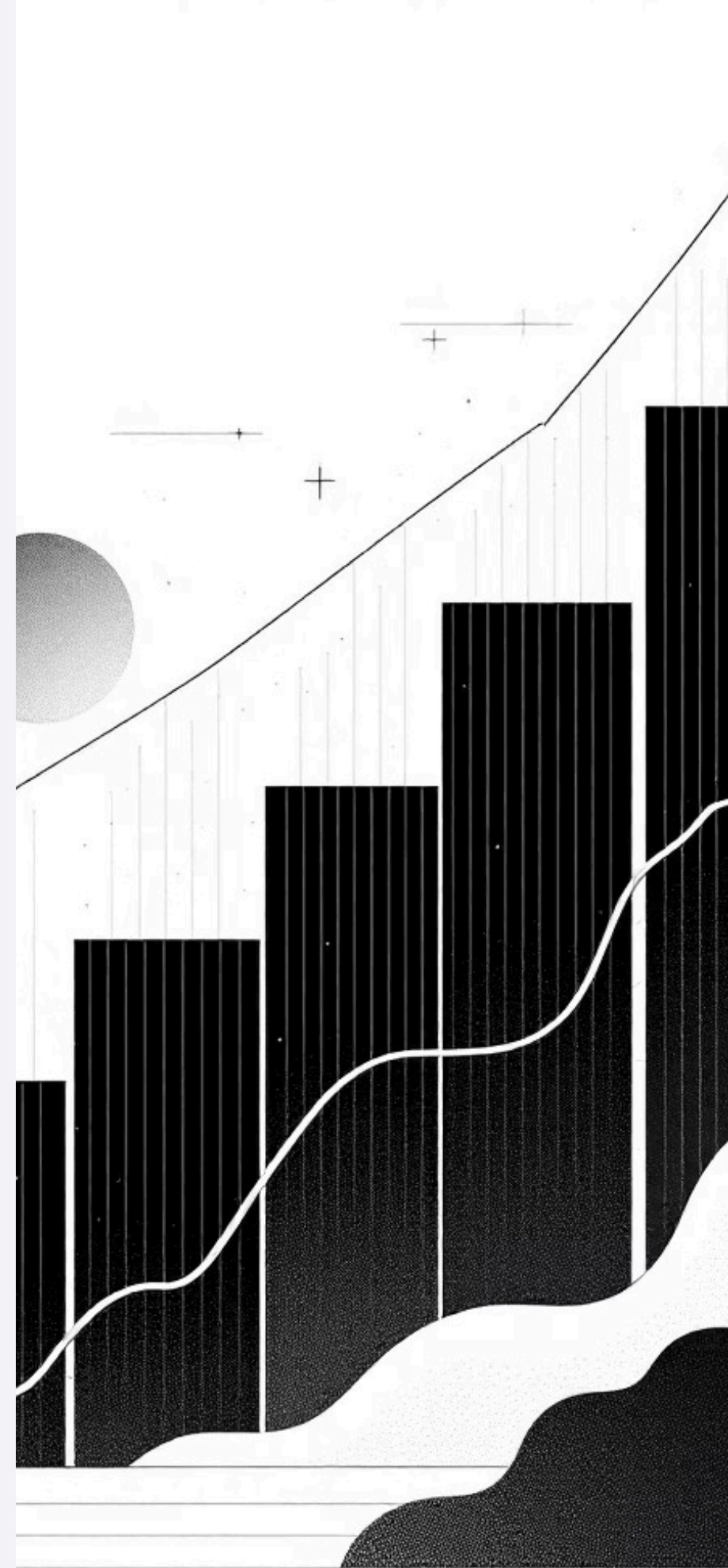
Process scales with organizational growth without proportional cost increases

- **Data-Driven Decisions**

Quantitative scoring enables analytics and continuous optimization

- **Competitive Advantage**

Intelligent automation positions organization as innovative employer



Technical Architecture



Resume Submission Layer

Form trigger and upload interface for candidate applications



Processing Layer

Orchestration engine manages workflow, text extraction, and normalization



Intelligence Layer (OpenAI GPT)

Skill extraction, experience analysis, JD matching, score calculation, JSON output



Decision Layer

Conditional logic applies passing score threshold for automated decisions



Data Layer

Database storage with audit logging and timestamp tracking



Communication Layer

Automated email workflows for shortlisted and rejected candidates

Ready for Implementation

Proceed with this strategic HR automation initiative to improve hiring speed, enhance decision accuracy, enable scalable recruitment, reduce operational costs, and standardize evaluation processes across your organization.